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United States Patent	12399467
Kind Code	B2
Date of Patent	August 26, 2025
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Building management systems and methods for tuning fault detection thresholds

Abstract

Building management systems and methods for autonomously tuning rule thresholds are disclosed. In one aspect, the method includes a providing a rule including a threshold, the rule used to determine whether building equipment has a fault. The method further includes receiving a state of the building equipment, assessing using a machine learning model whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determining a new threshold based on the assessment of the machine learning model, and replacing the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

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Appl. No.:	17/540725
Filed:	December 02, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230152755 A1	May. 18, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17529118 20211117 US 11934966 child-doc US 17540725

Publication Classification

Int. Cl.: G05B13/04 (20060101); G05B13/02 (20060101); G05B15/02 (20060101)

U.S. Cl.:

CPC **G05B13/0265** (20130101); G05B15/02 (20130101); G05B2219/2642 (20130101)

Field of Classification Search

USPC: 700/275

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) The present application is a continuation-in-part of U.S. patent application Ser. No. 17/529,118, filed Nov. 17, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) This application relates generally to a building system of a building. This application relates more particularly to systems for tuning fault detection of the building system.

(2) A building management system (BMS) is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof. Specifically, the present disclosure relates to a BMS with an equipment monitoring system to accurately determine whether the BMS is experiencing a fault.

SUMMARY

(3) One inventive aspect is a non-transitory computer-readable storage medium having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to provide a rule including a threshold, the rule used to determine whether building equipment has a fault. The instructions further cause the one or more processors to receive a state of the building equipment, assess, using a machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determine a new threshold based on the assessment of the machine learning model, and replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

(4) In some embodiments, the new threshold reduces a number of false positives a number of false negatives.

(5) In some embodiments, the instructions further cause the one or more processors to accumulate training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

(6) In some embodiments, the rule includes a condition portion including a comparison of the state to the threshold and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

(7) In some embodiments, the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

(8) In some embodiments, the instructions further cause the one or more processors to perturb the building equipment with multiple values of the threshold to provide additional data for the machine learning model.

(9) In some embodiments, the first machine learning model is configured to accurately predict false positives at a first rate, and the instructions further cause the one or more processors to retrain the first machine learning model when the first rate exceeds a first threshold rate.

(10) In some embodiments, the second machine learning model is configured accurately predict false negatives at a second rate, and the instructions further cause the one or more processors to retrain the second machine learning model when the second rate exceeds a second threshold rate.

- (11) In some embodiments, the one or more processors are configured to use a constrained nonlinear optimization to determine the new threshold.
- (12) Another aspect is a method including providing, by a processing circuit, a rule including a threshold, the rule used to determine whether building equipment has a fault. The method includes receiving, by the processing circuit, a state of the building equipment, assessing, by the processing circuit using a machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determining, by the processing circuit, a new threshold based on the assessment of the machine learning model, and replacing, by the processing circuit, the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.
- (13) In some embodiments, the new threshold reduces a number of false positives a number of false negatives.
- (14) In some embodiments, accumulating, by the processing circuit, training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.
- (15) In some embodiments, the rule includes a condition portion including a comparison of the state to the threshold and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.
- (16) In some embodiments, the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.
- (17) In some embodiments, perturbing, by the processing circuit, the building equipment with multiple values of the threshold to provide additional data for the machine learning model.
- (18) In some embodiments, the first machine learning model is configured to accurately predict false positives at a first rate, and the method further includes retraining, by the processing circuit, the first machine learning model when the first rate exceeds a first threshold rate.
- (19) In some embodiments, the second machine learning model is configured accurately predict false negatives at a second rate, and the method further includes retraining, by the processing circuit, the second machine learning model when the second rate exceeds a second threshold rate.
- (20) In some embodiments, the method further includes using, by the processing circuit, a constrained nonlinear optimization to determine the new threshold.
- (21) Another aspect is a building system including one or more storage devices storing instructions thereon and one or more processors which execute the instructions causing the one or more processors to provide a rule used to determine whether a building equipment has a fault, perturb the building equipment having a plurality of states with a plurality of corresponding thresholds for the rule for determining whether a fault exists, determine whether a fault exists based on the perturbed building equipment, and receive feedback of whether the determination of whether a fault exists is a false positive or a false negative, or a true positive. The one or more processors further execute the instructions causing the one or more processors to provide training data to a machine learning model, the training data including the plurality of states and the plurality of corresponding thresholds as inputs and the feedback of false positive or false negative as outputs. The one or more processors further execute the instructions causing the one or more processors to receive a current state of the building equipment and a current threshold of the rule, assess, using the trained machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the current state and the current threshold, determine a new threshold based on the assessment of the trained machine learning model, and replace the current threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.
- (22) In some embodiments, the instructions further cause the one or more processors to use constrained nonlinear optimization to find the new threshold that reduces a number of false positives a number of false negatives.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and

better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 is a drawing of a building equipped with a HVAC system, according to an exemplary embodiment.

(3) FIG. 2 is a block diagram of a waterside system that may be used in conjunction with the building of FIG. 1, according to an exemplary embodiment.

(4) FIG. 3 is a block diagram of an airside system that may be used in conjunction with the building of FIG. 1, according to an exemplary embodiment.

(5) FIG. 4 is a block diagram of a building automation system (BAS) that may be used to monitor and/or control the building of FIG. 1, according to an exemplary embodiment.

(6) FIG. 5 is a block diagram of a system including a fault detection system, according to some embodiments.

(7) FIG. 6 is a block diagram of a system to find an optimal threshold of a fault detection and diagnosis rule, according to an exemplary embodiment.

(8) FIG. 7 is a flow diagram of a process for identifying values for the parameters of the threshold of FIG. 6, according to an exemplary embodiment.

(9) FIG. 8 is an optimization algorithm that can be performed to identify the optimal threshold for the fault detection and diagnosis rule of FIGS. 6 and 7, according to an exemplary embodiment.

(10) FIG. 9 is a flow diagram of a process for calculating the optimal threshold, according to an exemplary embodiment.

DETAILED DESCRIPTION

Overview

(11) Referring generally to the FIGURES, systems and methods for autonomous fault detection and diagnosis rule threshold tuning are shown, according to various exemplary embodiments. One method of detecting faults in buildings and their equipment is using rules to detect and diagnose faults. For example, when a building controller detects that a rule is satisfied, the building controller may determine that a subsystem pertaining to or associated with the rule has or may have a fault that should be adjusted and/or fixed.

(12) A common technique of detecting faults in building equipment is using rules within building management systems (BMS). There may be one or more rules pertaining any particular building equipment (e.g., air handling unit, lighting equipment, etc.). Ideally the BMS will always correctly determine whether the building equipment is experiencing a fault or not. However, it is possible that the BMS determines that a fault exists even when the building equipment is operating normally (false positive), or that the BMS determines that a fault does not exist even when the building equipment does actually have a fault (false negative). When a fault is detected, the building manager may be notified by an alarm so that the building equipment may be inspected and/or a work order may be generated for a technician to fix the building equipment. This failure in accurate determination can lead to equipment failures, occupant discomfort and/or safety concerns, excessive costs, and waste.

(13) Rules often include a condition portion and an action portion. When the condition portion is satisfied, the action portion may be executed. For example, there may be a rule for an air handling unit (AHU) that states, in the condition portion, that the supply air temperature is below a first threshold temperature and the fan speed of the AHU is greater than a second threshold rotations per minute (rpm). The rule may have an action portion that states that the AHU is at fault. In this rule, the condition portion is satisfied if both the supply air temperature is below the first threshold and the fan speed is greater than the second threshold, and the system can receive a determination that the AHU is faulty.

(14) Rules are not always set for every condition and for every equipment. Accordingly, the rules may need to be adjust depending on operating conditions, equipment, occupants, etc. Further, the rules may need to be adjust because the equipment may degrade over time. If a threshold is too small, normal variation in operating conditions may result in false alarms, and if the threshold is too great, only a few extremely severe faults may be detected. Furthermore, manually adjusting or tuning thresholds is very difficult because of inaccuracies or overcompensation, which can lead to similar or different problems. Accordingly, there is a need to automatically tune the thresholds in rules.

(15) In the present disclosure, systems and methods of automatically tuning the thresholds in rules are described. First, some data needs to be collected concerning the rules and whether the rules resulted in false positives or false negatives. The data can include the state of the building equipment, the threshold of the rule, and a determination of whether the rule resulted in a false positive or a false negative. This data can be provided as training data for a machine learning model or multiple models that can be trained to predict when a false positive or a false negative is going to occur based on the state and threshold. Once the model is trained, an optimization technique can be used to determine the optimal threshold level that minimizes the numbers of false positives and false negatives. The optimal threshold can then replace the existing threshold so that false positives and false negatives are minimized.

(16) Building Management System Overview

(17) Referring now to FIGS. 1-4, an exemplary building automation system (BAS) and HVAC system in which the systems and methods of the present invention can be implemented are shown, according to an exemplary embodiment. Referring particularly to FIG. 1, a perspective view of a building 10 is shown. Building 10 is served by a BAS. A BAS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BAS can include, for example, a HVAC system, a security system, a lighting system, a fire alarming system, any other system that is capable of managing building functions or devices, or any combination thereof.

(18) The BAS that serves building 10 includes an HVAC system 100. HVAC system 100 can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building 10. For example, HVAC system 100 is shown to include a waterside system 120 and an airside system 130. Waterside system 120 can provide a heated or chilled fluid to an air handling unit of airside system 130. Airside system 130 can use the heated or chilled fluid to heat or cool an airflow provided to building 10. An exemplary waterside system and airside system which can be used in HVAC system 100 are described in greater detail with reference to FIGS. 2-3.

(19) HVAC system 100 is shown to include a chiller 102, a boiler 104, and a rooftop air handling unit (AHU) 106. Waterside system 120 can use boiler 104 and chiller 102 to heat or cool a working fluid (e.g., water, glycol, etc.) and can circulate the working fluid to AHU 106. In various embodiments, the HVAC devices of waterside system 120 can be located in or around building 10 (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler 104 or cooled in chiller 102, depending on whether heating or cooling is required in building 10. Boiler 104 can add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller 102 can place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller 102 and/or boiler 104 can be transported to AHU 106 via piping 108.

(20) AHU 106 can place the working fluid in a heat exchange relationship with an airflow passing through AHU 106 (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building 10, or a combination of both. AHU 106 can transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU 106 can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid can then return to chiller 102 or boiler 104 via piping 110.

(21) Airside system 130 can deliver the airflow supplied by AHU 106 (i.e., the supply airflow) to building 10 via air supply ducts 112 and can provide return air from building 10 to AHU 106 via air return ducts 114. In some embodiments, airside system 130 includes multiple variable air volume (VAV) units 116. For example, airside system 130 is shown to include a separate VAV unit 116 on each floor or zone of building 10. VAV units 116 can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building 10. In other embodiments, airside system 130 delivers the supply airflow into one or more zones of building 10 (e.g., via supply ducts 112) without using intermediate VAV units 116 or other flow control elements. AHU 106 can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU 106 can receive input from sensors located within AHU 106 and/or within the building zone and can adjust the flow rate, temperature, or other attributes of the supply airflow through AHU 106 to

achieve setpoint conditions for the building zone.

(22) Referring now to FIG. 2, a block diagram of a waterside system **200** is shown, according to an exemplary embodiment. In various embodiments, waterside system **200** can supplement or replace waterside system **120** in HVAC system **100** or can be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, waterside system **200** can include a subset of the HVAC devices in HVAC system **100** (e.g., boiler **104**, chiller **102**, pumps, valves, etc.) and can operate to supply a heated or chilled fluid to AHU **106**. The HVAC devices of waterside system **200** can be located within building **10** (e.g., as components of waterside system **120**) or at an offsite location such as a central plant.

(23) In FIG. 2, waterside system **200** is shown as a central plant having a plurality of subplants **202-212**. Subplants **202-212** are shown to include a heater subplant **202**, a heat recovery chiller subplant **204**, a chiller subplant **206**, a cooling tower subplant **208**, a hot thermal energy storage (TES) subplant **210**, and a cold thermal energy storage (TES) subplant **212**. Subplants **202-212** consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve the thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant **202** can be configured to heat water in a hot water loop **214** that circulates the hot water between heater subplant **202** and building **10**. Chiller subplant **206** can be configured to chill water in a cold water loop **216** that circulates the cold water between chiller subplant **206** and building **10**. Heat recovery chiller subplant **204** can be configured to transfer heat from cold water loop **216** to hot water loop **214** to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop **218** can absorb heat from the cold water in chiller subplant **206** and reject the absorbed heat in cooling tower subplant **208** or transfer the absorbed heat to hot water loop **214**. Hot TES subplant **210** and cold TES subplant **212** can store hot and cold thermal energy, respectively, for subsequent use.

(24) Hot water loop **214** and cold water loop **216** can deliver the heated and/or chilled water to air handlers located on the rooftop of building **10** (e.g., AHU **106**) or to individual floors or zones of building **10** (e.g., VAV units **116**). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air can be delivered to individual zones of building **10** to serve the thermal energy loads of building **10**. The water then returns to subplants **202-212** to receive further heating or cooling.

(25) Although subplants **202-212** are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) can be used in place of or in addition to water to serve the thermal energy loads. In other embodiments, subplants **202-212** can provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system **200** are within the teachings of the present invention.

(26) Each of subplants **202-212** can include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

(27) Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

(28) Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** can also include one or more pumps or valves configured to control the

flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** can also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

(29) In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**, **224**, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves can be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** can include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

(30) Referring now to FIG. **3**, a block diagram of an airside system **300** is shown, according to an exemplary embodiment. In various embodiments, airside system **300** can supplement or replace airside system **130** in HVAC system **100** or can be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** can include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and can be located in or around building **10**. Airside system **300** can operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

(31) In FIG. **3**, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, AHU **302** can receive return air **304** from building zone **306** via return air duct **308** and can deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. **1**) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** can be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** can be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

(32) Each of dampers **316-320** can be operated by an actuator. For example, exhaust air damper **316** can be operated by actuator **324**, mixing damper **318** can be operated by actuator **326**, and outside air damper **320** can be operated by actuator **328**. Actuators **324-328** can communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** can receive control signals from AHU controller **330** and can provide feedback signals to AHU controller **330**. Feedback signals can include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that can be collected, stored, or used by actuators **324-328**. AHU controller **330** can be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

(33) Still referring to FIG. **3**, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** can be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** can communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

(34) Cooling coil **334** can receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and can return the chilled fluid to waterside system **200** via piping **344**. Valve **346** can be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

(35) Heating coil **336** can receive a heated fluid from waterside system **200** (e.g., from hot water loop **214**) via piping **348** and can return the heated fluid to waterside system **200** via piping **350**. Valve **352** can be

positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

(36) Each of valves **346** and **352** can be controlled by an actuator. For example, valve **346** can be controlled by actuator **354** and valve **352** can be controlled by actuator **356**. Actuators **354-356** can communicate with AHU controller **330** via communications links **358-360**. Actuators **354-356** can receive control signals from AHU controller **330** and can provide feedback signals to controller **330**. In some embodiments, AHU controller **330** receives a measurement of the supply air temperature from a temperature sensor **362** positioned in supply air duct **312** (e.g., downstream of cooling coil **334** and/or heating coil **336**). AHU controller **330** can also receive a measurement of the temperature of building zone **306** from a temperature sensor **364** located in building zone **306**.

(37) In some embodiments, AHU controller **330** operates valves **346** and **352** via actuators **354-356** to modulate an amount of heating or cooling provided to supply air **310** (e.g., to achieve a setpoint temperature for supply air **310** or to maintain the temperature of supply air **310** within a setpoint temperature range). The positions of valves **346** and **352** affect the amount of heating or cooling provided to supply air **310** by cooling coil **334** or heating coil **336** and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU controller **330** can control the temperature of supply air **310** and/or building zone **306** by activating or deactivating coils **334-336**, adjusting a speed of fan **338**, or a combination of both.

(38) Still referring to FIG. **3**, airside system **300** is shown to include a building automation system (BAS) controller **366** and a client device **368**. BAS controller **366** can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system **300**, waterside system **200**, HVAC system **100**, and/or other controllable systems that serve building **10**. BAS controller **366** can communicate with multiple downstream building systems or subsystems (e.g., HVAC system **100**, a security system, a lighting system, waterside system **200**, etc.) via a communications link **370** according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller **330** and BAS controller **366** can be separate (as shown in FIG. **3**) or integrated. In an integrated implementation, AHU controller **330** can be a software module configured for execution by a processor of BAS controller **366**.

(39) In some embodiments, AHU controller **330** receives information from BAS controller **366** (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BAS controller **366** (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller **330** can provide BAS controller **366** with temperature measurements from temperature sensors **362-364**, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BAS controller **366** to monitor or control a variable state or condition within building zone **306**.

(40) Client device **368** can include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system **100**, its subsystems, and/or devices. Client device **368** can be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device **368** can be a stationary terminal or a mobile device. For example, client device **368** can be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. Client device **368** can communicate with BAS controller **366** and/or AHU controller **330** via communications link **372**.

(41) Referring now to FIG. **4**, a block diagram of a building automation system (BAS) **400** is shown, according to an exemplary embodiment. BAS **400** can be implemented in building **10** to automatically monitor and control various building functions. BAS **400** is shown to include BAS controller **366** and a plurality of building subsystems **428**. Building subsystems **428** are shown to include a building electrical subsystem **434**, an information communication technology (ICT) subsystem **436**, a security subsystem **438**, a HVAC subsystem **440**, a lighting subsystem **442**, a lift/escalators subsystem **432**, and a fire safety subsystem **430**. In various embodiments, building subsystems **428** can include fewer, additional, or

alternative subsystems. For example, building subsystems **428** can also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building **10**. In some embodiments, building subsystems **428** include waterside system **200** and/or airside system **300**, as described with reference to FIGS. 2-3.

(42) Each of building subsystems **428** can include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem **440** can include many of the same components as HVAC system **100**, as described with reference to FIGS. 1-3. For example, HVAC subsystem **440** can include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building **10**. Lighting subsystem **442** can include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem **438** can include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.

(43) Still referring to FIG. 4, BAS controller **366** is shown to include a communications interface **407** and a BAS interface **409**. Interface **407** can facilitate communications between BAS controller **366** and external applications (e.g., monitoring and reporting applications **422**, enterprise control applications **426**, remote systems and applications **444**, applications residing on client devices **448**, etc.) for allowing user control, monitoring, and adjustment to BAS controller **366** and/or subsystems **428**. Interface **407** can also facilitate communications between BAS controller **366** and client devices **448**. BAS interface **409** can facilitate communications between BAS controller **366** and building subsystems **428** (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

(44) Interfaces **407**, **409** can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems **428** or other external systems or devices. In various embodiments, communications via interfaces **407**, **409** can be direct (e.g., local wired or wireless communications) or via a communications network **446** (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces **407**, **409** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, interfaces **407**, **409** can include a Wi-Fi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces **407**, **409** can include cellular or mobile phone communications transceivers. In one embodiment, communications interface **407** is a power line communications interface and BAS interface **409** is an Ethernet interface. In other embodiments, both communications interface **407** and BAS interface **409** are Ethernet interfaces or are the same Ethernet interface.

(45) Still referring to FIG. 4, BAS controller **366** is shown to include a processing circuit **404** including a processor **406** and memory **408**. Processing circuit **404** can be communicably connected to BAS interface **409** and/or communications interface **407** such that processing circuit **404** and the various components thereof can send and receive data via interfaces **407**, **409**. Processor **406** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

(46) Memory **408** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** can be or include volatile memory or non-volatile memory. Memory **408** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an exemplary embodiment, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

(47) In some embodiments, BAS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BAS controller **366** can be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications **422** and **426** as existing outside of BAS controller **366**, in some embodiments, applications **422** and **426**

can be hosted within BAS controller **366** (e.g., within memory **408**).

(48) Still referring to FIG. **4**, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration layer **420**. Layers **410-420** can be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by each of layers **410-420** in BAS **400**.

(49) Enterprise integration layer **410** can be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** can be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control applications **426** can also or alternatively be configured to provide configuration GUIs for configuring BAS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at interface **407** and/or BAS interface **409**.

(50) Building subsystem integration layer **420** can be configured to manage communications between BAS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** can receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems **428**. Building subsystem integration layer **420** can also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

(51) Demand response layer **414** can be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization can be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** can receive inputs from other layers of BAS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.). The inputs received from other layers can include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs can also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

(52) According to an exemplary embodiment, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** can also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** can determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.

(53) In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models can include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models can represent collections of building equipment (e.g., subplants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

(54) Demand response layer **414** can further include or draw upon one or more demand response policy definitions (e.g., databases, XML files, etc.). The policy definitions can be edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs can be tailored for the user's application, desired comfort level, particular building equipment, or based on

other concerns. For example, the demand response policy definitions can specify which equipment can be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

(55) Integrated control layer **418** can be configured to use the data input or output of building subsystem integration layer **420** and/or demand response layer **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated supersystem. In an exemplary embodiment, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** can be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.

(56) Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** can be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration can reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer **418** can be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

(57) Integrated control layer **418** can be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints can also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** can be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

(58) Automated measurement and validation (AM&V) layer **412** can be configured to verify that control strategies commanded by integrated control layer **418** or demand response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** can be based on building system energy models and/or equipment models for individual BAS devices or subsystems. For example, AM&V layer **412** can compare a model-predicted output with an actual output from building subsystems **428** to determine an accuracy of the model.

(59) Fault detection and diagnostics (FDD) layer **416** can be configured to provide on-going fault detection for building subsystems **428**, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer **414** and integrated control layer **418**. FDD layer **416** can receive data inputs from integrated control layer **418**, directly from one or more building subsystems or devices, or from another data source. FDD layer **416** can automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults can include providing an alarm message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

(60) FDD layer **416** can be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer **420**. In other exemplary embodiments, FDD layer **416** is configured to provide “fault” events to integrated control layer **418** which executes control strategies and policies in response to the received fault events. According to an exemplary embodiment, FDD layer **416** (or a policy executed by an

integrated control engine or business rules engine) can shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

(61) FDD layer **416** can be configured to store or access a variety of different system data stores (or data points for live data). FDD layer **416** can use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems **428** can generate temporal (i.e., time-series) data indicating the performance of BAS **400** and the various components thereof. The data generated by building subsystems **428** can include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer **416** to expose when the system begins to degrade in performance and alarm a user to repair the fault before it becomes more severe.

(62) Autonomous FDD Rule Threshold Tuning

(63) Referring now to FIG. 5, a block diagram of a system **500** including a false positive and false negative prediction system **502** that is configured to predict false positives and false negatives of faults detected in building equipment by the FDD layer **416** in a building management system (e.g., BMS **400**) is shown, according to an exemplary embodiment. False positive and false negative prediction system **502** may operate in a cloud environment or locally by a processor at the building management system. False positive and false negative prediction system **502** may implement one or more machine learning models to predict false positives and false negatives of faults that are detected in the building equipment. False positive and false negative prediction system **502** may do so by inputting measurements of various points of the piece of building equipment into the machine learning models and determining whether individual output confidence scores for false positives and false negatives from the models satisfy a predetermined criteria (e.g., exceed a predetermined threshold, is the highest predicted confidence score, etc.).

(64) According to various example implementations of the present disclosure, the FDD layer **416** may determine that a building equipment has a fault based on the trigger or satisfaction of a rule. A false positive may indicate that the building equipment actually is not experiencing a fault, and the rule incorrectly determined that a fault has occurred. Furthermore, the FDD layer **416** may determine that a building equipment is not experiencing a fault because a state of the building did not satisfy the rule. A false negative may indicate that the building equipment is actually experiencing a fault, and the rule incorrectly determined that the fault did not occur. Although the false positive prediction model **528** is described in detail herein, similar description may apply for the false negative prediction model **530** except for false negatives instead of false positives.

(65) As used herein, “points” or “data points” refer to sensor inputs, control outputs, control values, and/or different characteristics of the inputs and/or outputs. “Points” and/or “data points” may refer to various data objects relating to the inputs and the outputs such as BACnet objects. The objects may represent and/or include a point and/or group of points. The object may include various properties for each of the points. For example, an analog input may be a particular point represented by an object with one or more properties describing the analog input and another property describing the sampling rate of the analog input. For example, in some embodiments, a point is a data representation associated with a component of a BMS, such as a camera, thermostat, controller, VAV box, RTU, valve, damper, chiller, boiler, AHU, supply fan, etc.

(66) System **500** may include a user presentation system **538**, a building controller **540**, and building equipment **542**. Building controller **540** may be similar to or the same as BMS controller **366**. False positive and false negative prediction system **502** may be a component of or be within building controller **540**. In some embodiments, false positive and false negative prediction system **502** operates in the cloud as one or more cloud servers. Components **502** and **538-542** may communicate over a network (e.g., a synchronous or asynchronous network).

(67) False positive and false negative prediction system **502** may include a processing circuit **504**, a processor **506**, and a memory **508**. Processing circuit **504**, processor **506**, and/or memory **508** can be the same as, or similar to, processing circuit **404**, processor **406**, and/or memory **408**, as described with reference to FIG. 4. Memory **508** may include a data pre-processor **510**, equipment models **512a-n**, a training manager **514**, a data post-processor **516**, and a measurement database **518**. Memory **508** may

include any number of components.

(68) Data pre-processor **510** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, data pre-processor **510** includes a data collector **522**, a vector generator **524**, and an FP/FN identifier **526**. Data collector **522** may be configured to collect data that corresponds to different pieces of building equipment (e.g., building equipment **542**). Data collector **522** can be configured to retrieve and/or collect building data from a building management system and store the building data in measurement database **518**, in some embodiments. Data collector **522** can be configured to collect data automatically or, in some embodiments, poll sensors associated with building equipment **542** to collect data at predetermined time intervals set by an administrator. In some embodiments, data collector **522** can further be configured to collect data upon detecting that a value changed by an amount exceeding a threshold. In some embodiments, data collector **522** is configured to collect building data upon receiving a request from an administrator. The administrator may make the request from a client device. The administrator can request building data associated with any time period and building device.

(69) In some embodiments, the data collector **522** may execute rules related to the building equipment and determine whether any rules have been satisfied. For example, a rule pertaining to the AHU may state that if the air supply temperature is less than a first threshold and if the fan speed is greater than a second threshold, the AHU has a fault. Accordingly, the data collector **522** may collect the measurements (e.g., air supply temperature and fan speed) from the measurement database **518** and determine whether any rule is satisfied such that a fault is determined to exist for the AHU. Then, the data collector **522** may provide the measurements, the thresholds, and the determination of whether there is a fault to the vector generator **524** so that a vector may be generated for the prediction models.

(70) Data collector **522** may be configured to tag each data point of the data with timestamps indicating when the data point was generated and/or when data collector **522** collected the data point from the sensors. In some embodiments, data collector **522** can also tag the data with a device identifier tag indicating the building device from which the building data was collected. Thus, data collector **522** may store the timestamped data in measurement database **518** as a timeseries corresponding to how the measured values changed over time.

(71) As described herein, timeseries can be a collection of values for a particular point (e.g., a discharge air temperature point of an air handling unit, a discharge air temperature, a supply fan status, a zone air temperature, a humidity, a pressure, etc.) generated at different times (e.g., at periodic intervals). The values may include or be associated with identifiers of the building devices with which the points are associated (e.g., an air handler, a VAV box, a controller, a chiller, a boiler, vents, dampers, etc.). Each timeseries can include a series of values for the same point and a timestamp for each of the data values. For example, a timeseries for a point provided by a temperature sensor (e.g., provided through local gateways) can include a series of temperature values measured by the temperature sensor and the corresponding times at which the temperature values were measured. An example of a timeseries which can be generated by data collector **522** is as follows: [<key, timestamp1, value1>, <key, timestamp2, value2>, <key, timestamp3, value3>]

where key is an identifier of the source of the raw data samples (e.g., timeseries ID, sensor ID, device ID, etc.), timestamp_i may identify the time at which the *i*th sample was collected, and value_i may indicate the value of the *i*th sample.

(72) Measurement database **518** may be a database configured to store building data associated with a building management system (e.g., BMS **400**). Measurement database **518** can be a graph database, MySQL, Oracle, Microsoft SQL, PostgreSQL, DB2, document store, search engine, device identifier-value store, etc. Measurement database **518** can be configured to hold data including any amount of values and can be made up of any number of components. The data can include various measurements and states (e.g., temperature readings, pressure readings, device state readings, blade speeds, etc.) associated with building equipment (e.g., AHUs, chillers, boilers, VAVs, fans, etc.) of the building management system. In some embodiments, the building data is tagged with timestamps indicating times and dates that the values of the building data were generated by devices (e.g., sensors) of the building management system or retrieved by data collector **522**. It should be understood that, in some embodiments, measurement database **518** or any other type of data or data structure described herein may be or include a data structure configured to store digital twins of the building, building equipment, building spaces, building

occupants/people, events, or any other entity of or related to the building. For example, measurement database **518** and/or other elements described herein may be implemented as a knowledge graph having nodes representing entities of the building and edges representing relationships between the entities. Example implementations of such digital twins and knowledge graphs as may be utilized in conjunction with the features of the present disclosure can be found in U.S. patent application Ser. No. 17/529,118, filed Nov. 18, 2021, which is incorporated herein by reference in its entirety.

(73) In some embodiments, measurement database **518** may store setpoint values for different points of the building management system. The stored setpoint values may be associated with a schedule indicating the times in which building equipment **542** will operate so points of the building managements system will reach the corresponding stored setpoints. For example, a setpoint schedule may indicate that a kitchen should be 70 degrees at 7 P.M. but 68 degrees at 3 P.M. Accordingly, a controller (e.g., building controller **540**) may control the building equipment of the building to cause the temperature point to reach the setpoint temperature at the corresponding times. Measurement database **518** may include schedules for setpoints of any point of the building to reach a desired level of comfort for the building's occupants.

(74) Vector generator **524** may be configured to generate a feature vector that is configured to be input into machine learning models of equipment models **512a-n** from measurement database **518**. For example, the feature vector may include a state of the building equipment and a threshold that is used in the rule. For example, for an AHU of the building, there may be a rule that states if supply air temperature is less than a certain temperature and the fan speed is greater than a certain rpm, there is a fault with the AHU. This may rule may be expressed as follows: if ($c_{sub.1} < \epsilon_{sub.1}$ & $c_{sub.2} > \epsilon_{sub.2}$), then AHU has fault $F_{sub.1}$ where $c_{sub.1}$ corresponds to the supply air temperature, $\epsilon_{sub.1}$ corresponds to a first threshold, $c_{sub.2}$ corresponds to the fan speed, $\epsilon_{sub.2}$ corresponds to a second threshold, and $F_{sub.1}$ corresponds to the fault ID. For a first machine learning model, the state of the AHU may include the supply air temperature and the first threshold, and for a second machine learning model, the state of the AHU may include the fan speed and the second threshold. These pairs may be provided as feature vectors to the respective machine learning models.

(75) Vector generator **524** may generate such feature vectors upon determining an event has occurred. An event may be or include a detection that a value associated with the piece of building equipment is above a threshold, a determination that a predetermined time interval has passed since vector generator **524** previously executed the machine learning model, receipt of a user input indicating to execute the machine learning model, receipt of a signal from another computing device indicating to execute the machine learning model, etc. Vector generator **524** may monitor various aspects of the building management system to identify such events and determine when the events occur. For example, vector generator **524** may keep track of the times in which vector generator **524** executes the machine learning model. Vector generator **524** may maintain an internal clock and identify when a predetermined (e.g., a pre-programmed) time period has passed since the last time vector generator **524** executed the machine learning model and determine the predetermined time period has passed. Vector generator **524** may identify an event as occurring upon determining the predetermined time period has passed.

(76) Upon determining an event has occurred, vector generator **524** may generate a feature vector. Vector generator **524** may generate the feature vector by identifying the piece of building equipment that is associated with the event (e.g., the piece of building equipment that has a stored association with the event) and retrieve data that corresponds to the piece of building equipment. Vector generator **524** may retrieve the data that is associated with attributes or points of the piece of building equipment based on a stored association between the values and the attributes or points. Vector generator **524** may retrieve data that is associated with values from within a pre-configured time frame of the event (e.g., values that are associated with timestamps from a time frame before and/or after the event) and generate a feature vector using the retrieved values. Vector generator **524** may retrieve values that were collected from sensors of the building and/or values of setpoints that are stored in memory (e.g., measurement database **518**).

(77) Upon generating the feature vector, vector generator **524** may identify the machine learning model that is associated with the piece of building equipment that is associated with the event. Vector generator **524** may identify the machine learning model from equipment models **512a-n** that each includes or is otherwise associated with a different false positive prediction model **528** and/or a false negative prediction model **530**. Each of equipment models **512a-n** may be a data representation of a different piece of building equipment within the building management system. The false positive prediction models and/or false

negative prediction models of each equipment model **512a-n** may be associated with a device identifier of the respective equipment model **512a-n**. Vector generator **524** may identify, responsive to determining the identified event, false positive prediction model **528** and false negative prediction model **530** are associated with the same or an identical device identifier. Upon identifying false positive prediction model **528**, vector generator **524** may apply the generated feature vector to false positive prediction model **528** and execute false positive prediction model **528**.

(78) False positive prediction model **528** may be a machine learning model (e.g., a neural network, a random forest, a support vector machine, etc.) configured to output confidence scores associated with a whether the satisfaction of a fault rule (e.g., determination that the building equipment has a fault) related to a building equipment was a false positive. False positive prediction model **528** may be configured to output confidence scores for fault determinations based on feature vectors that are generated by vector generator **524** based on data that corresponds to a particular piece of building equipment (e.g., the piece of building equipment that the equipment model represents). False positive prediction model **528** may output confidence scores for whether the determination that a fault has occurred in the building equipment is a false positive. FP/FN identifier **526** may identify the confidence scores and/or determine whether the determination of a fault in the piece of building equipment is a false positive in the future based on the confidence scores.

(79) FP/FN identifier **526** may be configured to use a predetermined criteria to determine if and/or when a fault is likely to occur in a piece of building equipment. The predetermined criteria may be a threshold and/or one or more rules. For instance, FP/FN identifier **526** may determine a fault determination is a false positive by comparing the confidence score to a predetermined threshold. Responsive to determining the score exceeds the threshold, FP/FN identifier **526** may determine that the fault determination (e.g., the fact that there was a fault in the building equipment) is a false positive. However, responsive to determining the score does not exceed the threshold, the data processing system may determine that the fault determination is not likely a false positive (e.g., the fault is actually a true positive). The data processing system may compare the confidence score to any rule or threshold.

(80) Upon determining a confidence score for a rule satisfied the predetermined criteria, FP/FN identifier **526** may identify the rule that was determined to have been satisfied and associated with the confidence score and an identification of the threshold in the rule that was used to determine the fault. In some embodiments, FP/FN identifier **526** may generate an alert indicating the fault did not occur in the building equipment and transmit the alert to a client device (e.g., an administrative device) so an administrator can view the alert and keep the building equipment operation/online. In some embodiments, FP/FN identifier **526** may feed the identification of the false positive back to vector generator **524**, which in turn can use the identification to generate a new feature vector.

(81) False negative prediction model **530** may be a machine learning model similar to false positive prediction model **528** that is configured to predict when false negatives occurred. False negative prediction model **530** may be configured to output confidence scores for fault determinations based on feature vectors that are generated by vector generator **524** based on data that corresponds to a particular piece of building equipment (e.g., the piece of building equipment that the equipment model represents). False positive prediction model **528** may output confidence scores for false positives and false negatives indicating likelihoods that the individual false positives and false negatives are the correct prediction.

(82) Threshold optimizer **542** may receive the output confidence scores and process the scores to optimize the threshold that was used to determine the faults in the rules that were used in the false positive prediction model **528** and the false negative prediction model **530**. The threshold optimizer **542** may use constrained nonlinear optimization to find the optimal threshold value (based on the accumulated data so far) that minimizes the number of false positives and false negatives.

(83) Data post-processor **516** may receive the output confidence scores and the optimal threshold values and process the scores and thresholds to transmit a signal to user presentation system **538** and/or building controller **540** to adjust the threshold of the rules. Data post-processor **516** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, data post-processor **516** includes a record generator **536**. Record generator **536** may receive the predicted confidence scores and/or optimal threshold value and generate a record (e.g., a file, document, table, listing, message, notification, etc.) including confidence scores and/or the threshold values. Upon generating the record, record generator **536** may transmit the record to user presentation

system **538** for display and/or building controller **540** to use to adjust operation or the configuration of building equipment **542** to avoid the false positive and false negative in the rule.

(84) False positive and false negative prediction system **502** can provide indications of recommendations of how to adjust the threshold values to user presentation system **538** and/or building controller **540**. In some embodiments, building controller **540** uses the expected recommendations to operate building equipment **542** (e.g., control environmental conditions of a building, cause generators to turn on or off, charge or discharge batteries, etc.). Further, user presentation system **538** can receive the indications and/or recommendations and cause a client device to display indications (e.g., graphical elements, charts, words, numbers, etc.) of the threshold values and/or recommendations. For example, user presentation system **538** may receive the rule and the optimal threshold value so that the user can modify the rule in the FDD layer **416**.

(85) In some embodiments, false positive and false negative prediction system **502** trains the prediction models of equipment models **512a-n** using training manager **514**. Training manager **514** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, training manager **514** includes a false positive prediction model trainer **532** and/or a false negative prediction model trainer **534**. False positive prediction model trainer **532** may be configured to train false positive prediction model **528** and other false positive prediction models of equipment models **512a-n** to predict when false positives occur when a fault is detected for pieces of building equipment. False positive prediction model trainer **532** may feed labeled training data including measurements associated with points of a particular piece of building equipment to the false positive prediction model associated with the piece of building equipment. The respective false positive prediction model may output confidence scores for thresholds and false positive prediction model trainer **532** may determine differences between the predicted outputs and the labels and use back-propagation techniques according to a loss function to adjust the false positive prediction model's weights and parameters proportional to the determined differences. False positive prediction model trainer **532** may repeat these steps for any number of fault prediction machine learning models to train the machine learning models to predict future false positives for individual pieces of building equipment.

(86) Similarly, false negative prediction model trainer **534** may be configured to train false negative prediction model **530** and other false negative prediction models of equipment models **512a-n**. False negative prediction model trainer **534** may feed measurement data and/or identifications of false negatives to obtain confidence scores for the threshold in rules used in determining faults in the building equipment. False negative prediction model trainer **534** may identify labels indicating the correct output, determine differences between the correct output and the respective false negative prediction model's output, and use back-propagation techniques according to a loss function to adjust the false negative prediction model's weights and parameters according to the determined differences. False negative prediction model trainer **534** may repeat these steps for any number of false negative prediction models to the machine learning models to predict false negatives for individual pieces of building equipment. In some embodiments, false negative prediction model trainer **534** may train a false negative prediction model in real-time. In such embodiments, false negative prediction model trainer **534** may feed measurement data and/or identifications of false negatives into a false negative prediction model to obtain confidence scores for false negatives in a rule.

(87) A user may input levels of accuracy (e.g., correct, incorrect, partially correct, etc.) of the recommendations and/or the predicted false positives or false negatives. False positive prediction model trainer **532** or false negative prediction model trainer **534** may identify the input levels of accuracy, determine differences between the predicted confidence scores and the input levels of accuracy, and use back-propagation techniques with the false negative prediction model that predicted the confidence scores for the false positives and false negatives according to a loss function based on the differences. Thus, false negative prediction model trainer **534** may train false negative prediction models in real-time or near real-time, in some embodiments.

(88) In some embodiments, training manager **514** may operate in a cloud server and be configured to use training data from multiple building management systems to train false positive prediction models and/or false negative prediction models. Training manager **514** may be configured to train individual machine learning models using training data that is associated with multiple pieces of building equipment (e.g., building equipment of the same type) until the machine learning models are accurate to a threshold, and

then deploy the machine learning models to the local building management system to be used to make predictions for individual pieces of building equipment (and be further trained based only on data associated with the piece of building equipment). This may be advantageous in building management systems that do not have enough training data to train machine learning models to make accurate predictions.

(89) In such embodiments, training manager **514** may be configured to train the machine learning models using a weighting policy. The weight policy may include weights that can be applied to different training data sets. The weights may correspond to different building management systems and may be determined based on how trustworthy an administrator has determined data from a building management system to be and/or based on whether the data originated at a building management system for which the models are being trained. Training manager **514** may use the weights by weighting the differences in a loss function so that training data that is associated with higher weights cause higher shifts in the weights or parameters of a machine learning model than training data that is associated with lower weights during training. Thus, training manager **514** may control the training to improve the accuracy and speed with which machine learning models are trained to be employed at individual building management systems.

(90) Referring now to FIG. **6**, a system **600** of a rule (or trigger rule) **602** of a building equipment where parameters of the trigger rule **602** are trained is shown, according to an exemplary embodiment. In some embodiments, the system **600** is similar to the system **400** or system **500**. In the example described in FIG. **6** below, a simple rule for an AHU is described for the sake of clarity and brevity, but embodiments are not limited thereto, and the disclosed technology may be implemented for a variety of other building equipment and/or rules. For example, the building equipment may be related to fire safety **430**, lifts/escalators **432**, electrical **434**, ICT **436**, security **438**, HVAC **440**, lighting **442** or any other building equipment within or around buildings. Further, the rules may be related to any rule that is implemented within the FDD layer **416**.

(91) The system **600** can perturb parameters, $\epsilon_{\text{sub.1}}$ of the trigger rule **602**. For example, the trigger rule **602** may include a rule that if supply air temperature is less than $\epsilon_{\text{sub.1}}^{\circ}\text{C.}$, the rule is triggered (or satisfied), and a corresponding action be performed. The corresponding action can be a determination that a fault exists with the AHU. The perturbation of the parameters can be increasing or decreasing the parameters in set amounts from existing values. The perturbation of the parameters can be selecting a space of values for the parameters and/or randomizing the parameters and/or parameter space.

(92) With the perturbed values for $\epsilon_{\text{sub.1}}$, the BMS controller **366** can simulate the state of the AHU for various temperatures within a range. The simulation can be performed by the false positive and false negative prediction system **502** via the models **512a-512n**. The output of the models **512a-512n** can be false positive or false negative.

(93) The system **600** can analyze the states produced by the building equipment operation **604** to determine states of the building equipment. For example, the air supply temperature and fan speed can be generated and collected for each state. Furthermore, the state can be any metric that is collected from any portion of the building equipment. For example, the state can include supply air temperature, fan speed, AHU operating mode (heating, cooling with outdoor air, etc.), outside air temperature, zone temperature, zone humidity, etc. Once the state is collected, various rules may be calculated related to the building equipment to determine whether a fault exists.

(94) The system **600** can receive feedback **606** from a user on whether the fault that was computed is a false positive, false negative, or true positive. As discussed above, false positive can imply that although a rule was satisfied to determine that a fault existed, there is no fault. False negative can imply that although the rule was not satisfied, and therefore no fault was detected, the building equipment in fact does have a fault. And a true positive can imply that the fault that was detected is true and no adjustment needs to be made to the rule.

(95) The system **600** can generate accumulated training data **608**. The accumulated training data can include the values of the parameters $\epsilon_{\text{sub.1}}$, the state of the building equipment for each value of the parameters, and the labeled data including whether the fault determination was a false positive, false negative, or true positive.

(96) The system **600** can generate neural networks **610** for predicting whether the fault determination was a false positive or a false negative based on the state s and parameters ϵ . The neural networks **610** can be trained by the system **600** based on the accumulated training data **608**.

(97) The system **600** can then determine optimal values for the parameters ϵ . The system **600** can search a space of potential values for ϵ that consider false positives and false negatives predicted by the trained neural network models **610**. The optimization can be the relation **800** shown in FIG. **8**. The optimization **612** performed by the system **600** can be a method of computing the optimal threshold of a trigger condition using the neural network models **610** of rewards and solving a constrained nonlinear optimization model. In some embodiments, the optimal values for the parameters found by the system **600** can be presented to a user for review and/or approval via a user interface, e.g., via the client device **368** or **448**.

(98) Referring now to FIG. **7**, a process **700** for identifying values for the parameters of the trigger rule **602** of FIG. **6** is shown, according to an exemplary embodiment. The process **700** can be performed by the system **600** and/or any component of the system **600**. Furthermore, the process **700** can be performed by any computing device described herein.

(99) In step **702**, the system **600** can perturb a building equipment with various values for thresholds ϵ . The result of the perturbed parameters can result in various states, s . The states can be states of the building equipment as described herein. The perturbations can result in pairs (s, ϵ) that can be used to determine feedback for false positive, false negative, or true positive.

(100) In step **704**, the system **600** can accumulate the pairs and corresponding feedback to create neural network models, e.g., the neural networks **610** based on the data determined in step **702**. The neural networks **610** can predict false positive and false negative rewards as a function of the state and the threshold, e.g., $FP=f(s, \epsilon)$ and $FN=f(s, \epsilon)$.

(101) In step **706**, the system **600** can determine a value for the parameter, ϵ that minimizes a relation, $(\alpha_{\text{sub.1}} \cdot \text{Math.FP} + \alpha_{\text{sub.2}} \cdot \text{Math.FN})$. The minimization is shown in relation **800** of FIG. **8**. The values of $\alpha_{\text{sub.1}}$ and $\alpha_{\text{sub.2}}$ can weigh the various rewards in the relation that is minimized, e.g., the false positive and false negative reward.

(102) In step **708**, the system **600** can periodically repeat the steps **702-706**. For example, the system **600** can repeat the steps at a defined time period (e.g., every day, week, month, etc.). In some embodiments, the retraining may occur if the rate of false positives and/or the rate of false negatives exceed an acceptable range set by the user. For example, if the user set the acceptable false positive rate to be 10% or lower, and the neural network model for false positives had a rate of 15%, the model may be retrained.

(103) Referring now to FIG. **9**, a process **900** of determining the optimal threshold for a rule (e.g., building equipment rule, trigger rule, or fault rule) is shown, according to an exemplary embodiment. The process **900** can be performed by the system **400**, **500**, or **600** and/or any of their components or a combination of the components. The process **900** can be performed by any computing device or processing circuit described herein. Furthermore, the process **900** may include additional steps or certain steps may be removed depending on embodiments.

(104) In step **902**, the system can provide a rule including a threshold. The rule can pertain to any that is used by the FDD layer **416** to determine whether a fault exists within the building equipment or not. For example, if there is a rule that states if AHU mode is mechanical cooling with 100% outdoor air and the magnitude of the difference between outside air temperature and mixed air temperature is greater than a threshold, the AHU is faulty.

(105) In step **904**, the system can receive a state of the building equipment. In some embodiments, the state can include any or all values that are used to calculate whether a rule is satisfied or not. In the above example, the AHU running in a mode with mechanical cooling with 100% outdoor air, the outside air temperature, and mixed air temperature can all be part of the state of the AHU. Further, the threshold can be the value that is being optimized so that the system can reduce or minimize the number of false positives and false negatives.

(106) In step **906**, the system can assess, using a machine learning model, whether the determination of whether the building equipment has fault is a false positive or a false negative based on the state and the threshold. For example, if the above rule was satisfied, the system may determine that a fault exists in the AHU. However, in reality, the AHU may be operating normally and not have a fault as determined by occupants of the building or the building manager. Accordingly, the model's administrator may provide feedback to the system indicating that the fault is a false positive. In another example, the rule may not be satisfied and the system may then determine that the AHU does not have a fault. However, based on occupant or building manager feedback, it may be determined that the AHU is actually faulty, and the

administrator may provide feedback to the system that there was a false negative.

(107) In step **908**, the system may determine a new threshold based on the assessment of the machine learning model. This may be done using constrained nonlinear optimization that solves for the minimal number of false positives and false negatives based on the feedback received in step **906**. The new threshold may be higher or lower than the existing one, depending on the parameters set with the optimization equations. Further, the user may desire to reduce the number of false positives more than the false negatives, in which case the optimization parameters used in the equations may be adjusted so that greater weight is given to minimizing the false positives. In other embodiments, the user may be interested in reducing the false negatives more than the false positives, in which case greater weight can be given to reducing the false negatives more than the false positives.

(108) In step **910**, the system may replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault. The number of false positives and false negatives may be reduced and/or minimized.

(109) Configuration of Exemplary Embodiments

(110) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

(111) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(112) Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

(113) In various implementations, the steps and operations described herein may be performed on one processor or in a combination of two or more processors. For example, in some implementations, the various operations could be performed in a central server or set of central servers configured to receive data from one or more devices (e.g., edge computing devices/controllers) and perform the operations. In

some implementations, the operations may be performed by one or more local controllers or computing devices (e.g., edge devices), such as controllers dedicated to and/or located within a particular building or portion of a building. In some implementations, the operations may be performed by a combination of one or more central or offsite computing devices/servers and one or more local controllers/computing devices. All such implementations are contemplated within the scope of the present disclosure. Further, unless otherwise indicated, when the present disclosure refers to one or more computer-readable storage media and/or one or more controllers, such computer-readable storage media and/or one or more controllers may be implemented as one or more central servers, one or more local controllers or computing devices (e.g., edge devices), any combination thereof, or any other combination of storage media and/or controllers regardless of the location of such devices.

Claims

1. A non-transitory computer-readable storage medium having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to: provide a rule including a threshold, wherein the rule is used to generate a determination of whether building equipment has a fault; receive a state of the building equipment; perturb the building equipment with multiple values of the threshold to provide additional data for a machine learning model; generate, by the machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold; determine a new threshold based on the assessment of the machine learning model; and replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.
2. The non-transitory computer-readable storage medium of claim 1, wherein the new threshold reduces a number of false positives a number of false negatives.
3. The non-transitory computer-readable storage medium of claim 1, wherein the instructions further cause the one or more processors to accumulate training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.
4. The non-transitory computer-readable storage medium of claim 1, wherein the rule includes: a condition portion including a comparison of the state to the threshold; and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.
5. The non-transitory computer-readable storage medium of claim 4, wherein the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.
6. The non-transitory computer-readable storage medium of claim 5, wherein the first machine learning model is configured to accurately predict false positives at a first rate, and wherein the instructions further cause the one or more processors to retrain the first machine learning model when the first rate exceeds a first threshold rate.
7. The non-transitory computer-readable storage medium of claim 5, wherein the second machine learning model is configured accurately predict false negatives at a second rate, and wherein the instructions further cause the one or more processors to retrain the second machine learning model when the second rate exceeds a second threshold rate.
8. The non-transitory computer-readable storage medium of claim 1, wherein the one or more processors are configured to use a constrained nonlinear optimization to determine the new threshold.
9. A method comprising: providing, by a processing circuit, a rule including a threshold, wherein the rule is used to generate a determination of whether building equipment has a fault; receiving, by the processing circuit, a state of the building equipment; perturbing, by the processing circuit, the building equipment with multiple values of the threshold to provide additional data for a machine learning model; generating, by the processing circuit and the machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold; determining, by the processing circuit, a new threshold based on the assessment of the machine learning model; and replacing, by the processing circuit, the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

10. The method of claim 9, wherein the new threshold reduces a number of false positives a number of false negatives.

11. The method of claim 9, wherein accumulating, by the processing circuit, training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

12. The method of claim 9, wherein the rule includes: a condition portion including a comparison of the state to the threshold; and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

13. The method of claim 12, wherein the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

14. The method of claim 13, wherein the first machine learning model is configured to accurately predict false positives at a first rate, and wherein the method further comprises retraining, by the processing circuit, the first machine learning model when the first rate exceeds a first threshold rate.

15. The method of claim 13, wherein the second machine learning model is configured accurately predict false negatives at a second rate, and wherein the method further comprises retraining, by the processing circuit, the second machine learning model when the second rate exceeds a second threshold rate.

16. The method of claim 9, further comprising using, by the processing circuit, a constrained nonlinear optimization to determine the new threshold.

17. A building system comprising: one or more storage devices storing instructions thereon; and one or more processors, wherein the one or more processors execute the instructions causing the one or more processors to: provide a rule used to generate a determination of whether a building equipment has a fault; perturb the building equipment having a plurality of states with a plurality of corresponding thresholds for the rule for determining whether a fault exists; determine whether a fault exists based on the perturbed building equipment; receive feedback of whether the determination of whether a fault exists is a false positive or a false negative, or a true positive; provide training data to a machine learning model, wherein the training data includes the plurality of states and the plurality of corresponding thresholds as inputs and the feedback of false positive or false negative as outputs; receive a current state of the building equipment and a current threshold of the rule; generate, by the trained machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the current state and the current threshold; determine a new threshold based on the assessment of the trained machine learning model; and replace the current threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

18. The building system of claim 17, wherein the instructions further cause the one or more processors to use constrained nonlinear optimization to find the new threshold that reduces a number of false positives a number of false negatives.
